

Timing Analysis of Mux using OpenSTA

INTRODUCTION:

In this lab, students will learn how to perform static timing analysis (STA) on a synthesized 2x1 multiplexer using OpenSTA. The lab focuses on understanding timing behaviour of combinational logic by importing a gate-level netlist generated using Yosys, loading a standard cell library, applying timing constraints, and analysing delay paths. This experiment demonstrates how timing is verified for simple combinational circuits at the signoff level.

APPLICATION:

A 2x1 multiplexer is a fundamental combinational circuit used in digital systems to select one of two input signals based on a control signal. Multiplexers are widely used in data routing, bus selection, ALU input selection, and control logic. They form the backbone of complex digital systems such as processors, communication hardware, and embedded systems by enabling efficient data selection and resource sharing.

OBJECTIVE:

After completing this lab, you will be able to:

Import a synthesized gate-level netlist and standard cell library into OpenSTA (OSU018 lib).

- Apply input and output timing constraints using SDC format.
- Perform static timing analysis on combinational logic.
- Analyse propagation delay, slack, and critical timing paths.
- Interpret timing reports to evaluate performance of combinational circuits.

PROCEDURE TO CREATE A LAB:

Step 1: Set up Project Directory

- 1.1: Create a directory for your lab: `mkdir <project_name>`
- 1.2: Open lab directory: `cd <project_name>`
- 1.3: Create directory for project: `mkdir mux_2x1`
- 1.4: Open project directory: `cd mux_2x1`

Step 2: Create Design file

2.1: Open gvim editor for design with command: `gvim mux_2x1.v`

RTL Design Code:

```
module mux_2x1 (  
    input a,  
    input b,  
    input sel,  
    output out  
);  
assign out = sel ? b : a;  
endmodule
```

2.2: save the file using command- :wq

Step 3: Create .sdc file

3.1: Open gvim editor for the SDC with command: `gvim mux_2x1.sdc`

For 10ns:

```
create_clock -period 10 -name clk [get_ports clk]  
set_input_delay 5 -min -rise [get_ports clk] -clock clk  
set_input_delay 5 -max -fall [get_ports {a b sel}] -clock clk  
set_input_transition 5 -min -rise [get_ports {a b sel}] -clock clk  
set_input_transition 5 -max -fall [get_ports clk] -clock clk  
set_load -pin_load 4 [get_ports out]  
set_output_delay 2 -min -rise [get_ports out] -clock clk
```

Note: For 5ns, change the clock period to 5ns in .sdc file

Step 4: Static Timing Analysis for the design with OpenSTA

Invoke OpenSTA in the terminal using the command: `sta`

4:1 Use the following commands:

Loads the liberty to osu018 standard cell library

i) `read_liberty osu018_stdcells.lib`

Reads your synthesized gate-level Verilog netlist

ii) `read_verilog mux_2x1_synth.v`

Links the loaded netlist and library to the top module name

iii) link_design mux_2x1

Loads your timing constraints from an SDC file

iv) read_sdc mux_2x1.sdc

Runs timing analysis to check all timing constraints and generates a report.

v) report_checks

OBSERVATION:

- Analyse input-to-output propagation delay of the multiplexer
- Identify the critical timing path through selection logic
- Verify that no setup or hold violations occur
- Observe positive slack values indicating timing closure

OUTPUT:

Reports:

For 10ns:

→ for setup

```
OpenSTA> report_checks -path_delay max
Startpoint: sel (input port clocked by clk)
Endpoint: out (output port clocked by clk)
Path Group: clk
Path Type: max
```

Delay	Time	Description
0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
5.00	5.00	v input external delay
0.00	5.00	v sel (in)
86.78	91.78	^ _4_/Y (0AI21X1)
0.00	91.78	^ out (out)
	91.78	data arrival time
100.00	100.00	clock clk (rise edge)
0.00	100.00	clock network delay (ideal)
0.00	100.00	clock reconvergence pessimism
-5.00	95.00	output external delay
	95.00	data required time
	95.00	data required time
	-91.78	data arrival time
	3.22	slack (MET)

→ for hold

```
OpenSTA> report_checks -path_delay min
Startpoint: a (input port clocked by clk)
Endpoint: out (output port clocked by clk)
Path Group: clk
Path Type: min
```

Delay	Time	Description
0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
0.00	0.00	^ input external delay
0.00	0.00	^ a (in)
0.24	0.24	v _2_/Y (INVX1)
8.31	8.55	^ _4_/Y (OAI21X1)
0.00	8.55	^ out (out)
	8.55	data arrival time
0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
0.00	0.00	clock reconvergence pessimism
-2.00	-2.00	output external delay
	-2.00	data required time
	-2.00	data required time
	-8.55	data arrival time
	10.55	slack (MET)

For 5ns:

→ for setup

```
OpenSTA> report_checks -path_delay max
Startpoint: sel (input port clocked by clk)
Endpoint: out (output port clocked by clk)
Path Group: clk
Path Type: max
```

Delay	Time	Description
0.00	0.00	clock clk (rise edge)
0.00	0.00	clock network delay (ideal)
5.00	5.00	v input external delay
0.00	5.00	v sel (in)
86.78	91.78	^ _4_/Y (OAI21X1)
0.00	91.78	^ out (out)
	91.78	data arrival time
50.00	50.00	clock clk (rise edge)
0.00	50.00	clock network delay (ideal)
0.00	50.00	clock reconvergence pessimism
-5.00	45.00	output external delay
	45.00	data required time
	45.00	data required time
	-91.78	data arrival time
	-46.78	slack (VIOLATED)

→ for hold

```
OpenSTA> report_checks -path_delay min
Startpoint: a (input port clocked by clk)
Endpoint: out (output port clocked by clk)
Path Group: clk
Path Type: min

Delay      Time      Description
-----
0.00       0.00      clock clk (rise edge)
0.00       0.00      clock network delay (ideal)
0.00       0.00      ^ input external delay
0.00       0.00      ^ a (in)
0.24       0.24      v _2_/Y (INVX1)
8.31       8.55      ^ _4_/Y (OAI21X1)
0.00       8.55      ^ out (out)
0.00       8.55      data arrival time

0.00       0.00      clock clk (rise edge)
0.00       0.00      clock network delay (ideal)
0.00       0.00      clock reconvergence pessimism
-2.00      -2.00      output external delay
-2.00      -2.00      data required time

-2.00      -2.00      data required time
-8.55      -8.55      data arrival time

10.55      10.55      slack (MET)
```

COMPARISION:

<u>Timing</u>	<u>Setup</u>	<u>Hold</u>
10ns	3.22	10.55
5ns	-46.78	10.55

Note: Timing can be met by changing constraints in sdc.

CONCLUSION:

Static timing analysis of the 2x1 multiplexer using OpenSTA confirms that the synthesized combinational design meets all specified timing constraints. The timing reports show positive slack, indicating reliable signal propagation from inputs to output within the defined delay limits. This lab demonstrates that even simple combinational blocks must be timing-verified to ensure correct operation when integrated into larger digital systems.